

INTRODUCTION TO DATABASE

Project: Fruit Buisness

Course instructor: MD SAJID BIN-FAISAL
Department: CSE
Section: E
E-mail: sajid@aiub.edu

Group Member	ID
1.Alrazi Hosen Rizvy	22-47724-2
2.SK.Shahed Ali	22-47756-2
3.Farzana Zaman	22-48258-2
4. Tahmina Jerin Tinny	22-47595-2

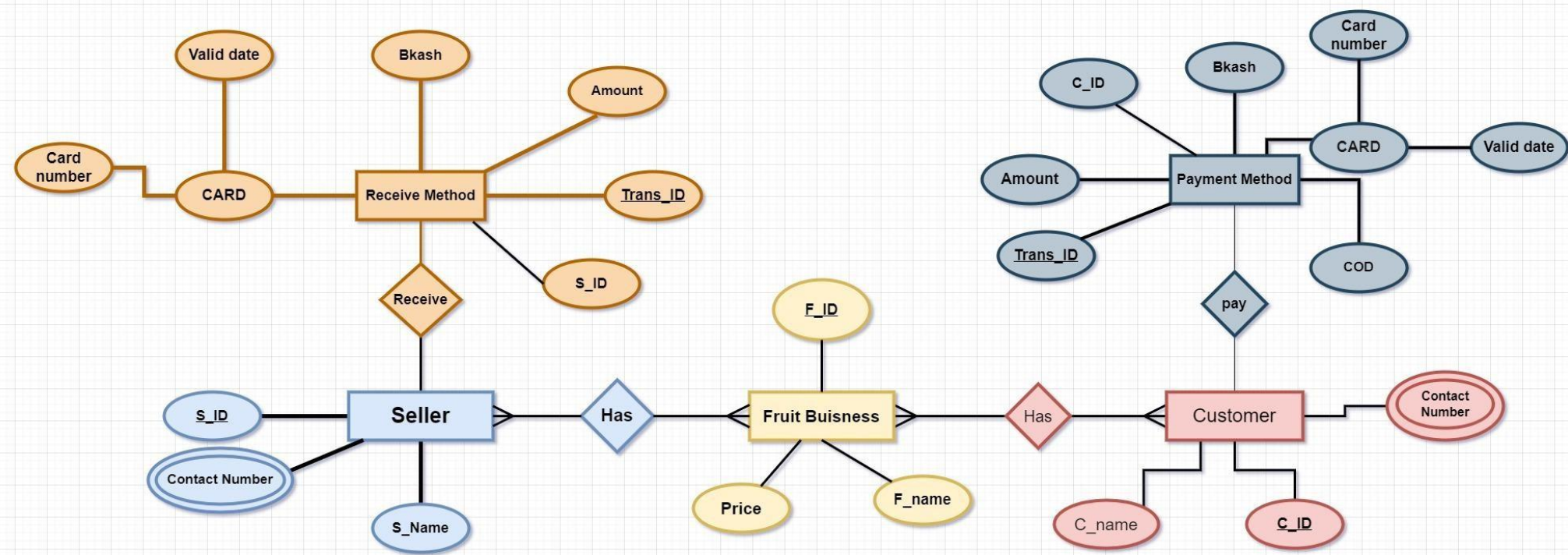
CONTAIN LIST

- ☐ Scenario of ER Diagram
- ☐ ER Diagram
- ☐ Normalization
- ☐ Table Creation
- ☐ Data Insertion
- ☐ Constraint
- ☐ Joinning
- ☐ Subquery
- ☐ View

CASE SCENARIO:

A system is decorated for fruit businesses to make people healthy and happy. The fruit business has two sides. One is the seller & other is customer. In this scenario, SCENARIO: A system is decorated for fruit businesses to make people healthy and happy. The fruit business has two sides. One is the seller & other is customer. In this scenario, Seller, Fruit Business, Customer, Receive Method, Payment Method. The entity set seller has attributes of s_id, Contact Number, and s_Name. The customer must have c_name, c_id, and contact number. Customers must pay for buying fruits. There is one method for this condition is a payment method in which the total attributes are trans_ID, amount, Bkash, valid date, and card number. Business means a way of a platform of economic condition. The attributes are fruit business platform f_id, price, and f_name.seller is linked strongly with customers to the transaction system. The seller must be in relation with receive circumstances method to survive with the business., Fruit Business, Customer, ReceiveMethod, Payment Method. The entity set seller has attributes of s_id, Contact Number, and s_Name. The customer must have c_name,c_id, and contact number. Customers must pay for buying fruits. There is one method for this condition payment method which total attributes are trans_ID, amount, Bkash, valid date, and card number. Business means a way of a platform of economic conditions. The attributes are fruit business platform f_id, price, and f_name. Sellers are linking strongly with customers to the transaction system. The seller must be in relation with receive circumstances method to survive with the business.

ER Diagram:



NORMALIZATION

FRUIT BUSINESS HAS SELLER

UNF:

- 1st: SID, Contact Number, S_Name, FID, Price, F_Name

1NF:

- 1st: SID, Contact Number, S_Name, FID, Price, F_Name

2NF:

- 1st: Fid, F_Name, Price
- 2nd: SID, S_Name
- 3rd: SID, contactNumber
- 4th: contactNumber, S_name
- 5th: SID, FID

3NF:

- 1st: Fid, F_Name
- 2nd: SID, S_Name
- 3rd: SID, contactNumber
- 4th: contactNumber, S_name
- 5th: SID, FID
- 6th: FID, Price

FRUIT BUSINESS HAS CUSTOMER:

UNF:

- 1st: FID, Price, F_Name, C_name, C_ID, Contact Number

1NF:

- 1st: FID, Price, F_Name, C_name, CID, Contact Number

2NF:

- 1st: CID, Contact Number
- 2nd: C_name, CID
- 3rd: Contact Number, C_name
- 4th: FID, Price, F_Name
- 5th: FID, CID

3NF:

- 1st: CID, Contact Number
- 2nd: C_name, CID
- 3rd: Contact Number, C_name
- 4th: FID, Price
- 5th: FID, F_Name
- 6th: FID, CID

NORMALIZATION

CUSTOMER PAY PAYMENT METHOD:

UNF:

- 1st: C_name, C_ID, Contact Number, Trans_ID, Amount, Bkash, Card number, Valid date

1NF:

- 1st: C_name, C_ID, Contact Number, Trans_ID, Amount, Bkash, Card number, Valid date

2NF:

- 1st : CID , Contact Number
- 2nd: C_name, CID
- 3rd: Contact Number , C_name
- 4th: TransId , Amount, Card Number, valid date, COD
- 5th: CID , TransId

3NF:

- 1st : CID , Contact Number
- 2nd: C_name, CID
- 3rd: Contact Number , C_name
- 4th: TransId , Amount
- 5th: TransId , COD
- 6th: Card Number, valid date
- 7th: CID , TransId

SELLER RECEIVE RECEIVE METHOD:

UNF:

- 1st: SID, Contact Number, S_Name, Trans_ID , Amount, Bkash, Card number, Valid date

1NF:

- 1st: SID, Contact Number , S_Name, Trans_ID , Amount, Bkash, Card number, Valid date

2NF:

- 1st : SID , Contact Number
- 2nd: SID , S_name
- 3rd: Contact Number , S_name
- 4th: TransId , Amount, Card Number, valid date, COD
- 5th: CID , TransId

3NF:

- 1st : SID , Contact Number
- 2nd: SID , S_name
- 3rd: Contact Number , S_name
- 4th: TransId , Amount
- 5th: TransId , COD
- 6th: Card Number, valid date
- 7th: CID , TransId

NORMALIZATION

FINAL TABLE:

1st: Fid, F_Name

2nd: SID, S_Name

3rd: SID, contactNumber

4th: contactNumber, S_name

5th: SID, FID

6th: FID, Price

7th: CID, Contact Number

8th: C_name, CID

9th: Contact Number, C_name

10th: FID, CID

11th: TransId, Amount

12th: TransId, COD

13th: Card Number, valid date

14th: CID, TransId

Table Creation:

Fruit Buisness

create table Fruit_buisness (F_ID number(2) primary key, F_name varchar2(32),
Price number(11));

The screenshot shows the Oracle Database Express Edition web interface. The browser tabs include 'SQL Commands', 'Bangla Database Tutorial 45: VIE', 'Untitled Diagram.drawio - diagra', 'Sign in - GitLab', 'have has relation in diagram - Yo', and 'receive - Google Search'. The address bar shows '127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::'. The page title is 'ORACLE Database Express Edition'. The user is 'SYS'. The breadcrumb is 'Home > SQL > SQL Commands'. The 'Autocommit' checkbox is checked, and the 'Display' dropdown is set to '100'. The 'Save' and 'Run' buttons are visible. The SQL command entered is:
`create table Fruit_buisness (F_ID number(2) primary key, F_name varchar2(32), Price number(11));`
`describe fruit_buisness;`
The 'Results' tab is selected, showing the table structure for 'FRUIT_BUISNESS'. The table has three columns: F_ID (Number, 2, 0, Primary Key), F_NAME (Varchar2, 32, -), and PRICE (Number, 11, 0). The 'Comment' column is empty. The 'Language' is set to 'en-us'. The footer shows 'Application Express 2.1.0.0.39' and 'Copyright © 1999, 2006, Oracle. All rights reserved.'.

SQL Commands

127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::

ORACLE Database Express Edition

User: SYS

Home > SQL > SQL Commands

☒ Autocommit Display 100 Save Run

```
create table Fruit_buisness (F_ID number(2) primary key, F_name varchar2(32), Price number(11));  
describe fruit_buisness;
```

Results Explain Describe Saved SQL History

Object Type TABLE Object FRUIT_BUISNESS

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FRUIT_BUISNESS	F_ID	Number	-	2	0	1	-	-	-
	F_NAME	Varchar2	32	-	-	-	✓	-	-
	PRICE	Number	-	11	0	-	✓	-	-

1 - 3

Language: en-us

Application Express 2.1.0.0.39
Copyright © 1999, 2006, Oracle. All rights reserved.

23°C Haze 10:40 PM

Customer

create table Customer (C_ID number(2) primary key, C_name varchar2(32),
Contact_Number number(11));

The screenshot shows the Oracle Database Express Edition web interface. The browser tabs include 'SQL Commands', 'Bangla Database Tutorial 45 : VIE', 'Untitled Diagram.drawio - diagr...', 'Sign in - GitLab', 'have has relation in diagram - Yo', and 'receive - Google Search'. The address bar shows '127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::'. The page title is 'ORACLE Database Express Edition'. The user is 'SYS'. The breadcrumb is 'Home > SQL > SQL Commands'. The 'Autocommit' checkbox is checked, and the 'Display' dropdown is set to '100'. The 'Save' and 'Run' buttons are visible. The SQL command entered is:
`create table Customer (C_ID number(2) primary key, C_name varchar2(32), Contact_Number number(11));
describe Customer;`
The 'Results' tab is selected, showing the table structure for 'CUSTOMER'. The table has three columns: C_ID, C_NAME, and CONTACT_NUMBER. The 'C_ID' column is the primary key. The 'C_NAME' column is nullable. The 'CONTACT_NUMBER' column is nullable. The table is located in the '1 - 3' schema.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUSTOMER	C_ID	Number	-	2	0	1	-	-	-
	C_NAME	Varchar2	32	-	-	-	✓	-	-
	CONTACT_NUMBER	Number	-	11	0	-	✓	-	-
1 - 3									

Application Express 2.1.0.00.39
Copyright © 1999, 2006, Oracle. All rights reserved.

Language: en-us

Seller

create table Seller(S_ID number(2) primary key, S_name varchar2(32),
Contact_Number number(11));

The screenshot shows the Oracle Database Express Edition web interface. The browser tabs include 'SQL Commands', 'Bangla Database Tutorial 45 : VIE', 'Untitled Diagram.drawio - diagra', 'Sign in · GitLab', and 'have has relation in diagram - Yo'. The address bar shows '127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::'. The page title is 'ORACLE Database Express Edition'. The user is 'SYS'. The breadcrumb is 'Home > SQL > SQL Commands'. The 'Autocommit' checkbox is checked, and the 'Display' dropdown is set to '100'. The SQL command entered is 'create table Seller (S_ID number(2) primary key, S_name varchar2(32), Contact_Number number(11));' followed by 'describe seller;'. The 'Save' and 'Run' buttons are visible. The 'Results' tab is selected, showing the table structure for 'SELLER'.

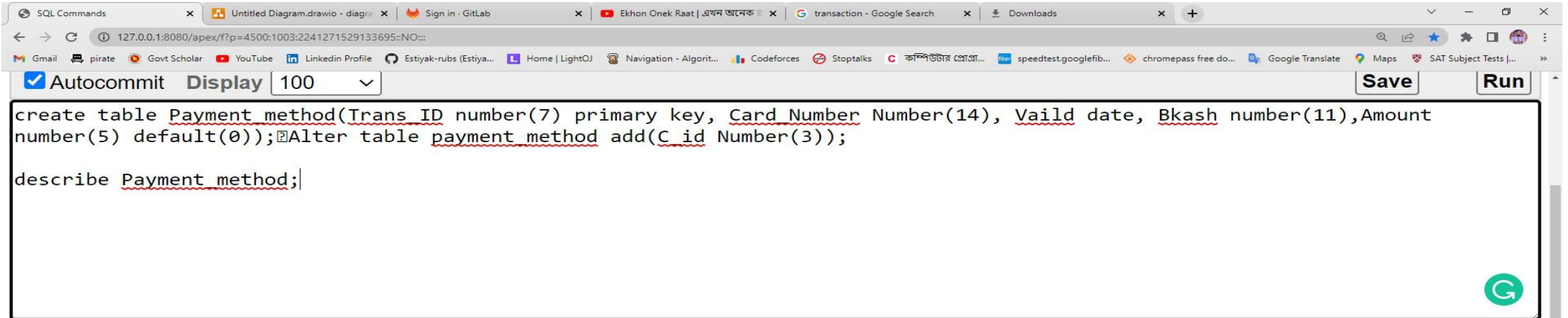
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SELLER	S_ID	Number	-	2	0	1	-	-	-
	S_NAME	Varchar2	32	-	-	-	✓	-	-
	CONTACT_NUMBER	Number	-	11	0	-	✓	-	-
1 - 3									

Application Express 2.1.0.00.39
Copyright © 1999, 2006, Oracle. All rights reserved.

Language: en-us

Payment Method

create table Payment_method(Trans_ID number(7) primary key, Card_Number Number(14), Vaild date, Bkash number(11),Amount number(5) default(0));
Alter table payment_method add(C_id Number(3));



The screenshot shows a web-based SQL IDE. The top navigation bar includes tabs for 'SQL Commands', 'Untitled Diagram.drawio - diagr...', 'Sign in - GitLab', 'Ekhn Onek Raat | এখন অনেক...', 'transaction - Google Search', and 'Downloads'. The address bar shows the URL '127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO...'. Below the address bar, there are links for 'Autocommit' (checked), 'Display' (set to 100), 'Save', and 'Run'. The main text area contains the following SQL commands:

```
create table Payment method(Trans ID number(7) primary key, Card Number Number(14), Vaild date, Bkash number(11),Amount  
number(5) default(0));Alter table payment method add(C_id Number(3));  
  
describe Payment_method;
```

A green circular 'Run' button is located at the bottom right of the text area.

Results Explain Describe Saved SQL History

Object Type TABLE Object PAYMENT_METHOD

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT_METHOD	TRANS_ID	Number	-	7	0	1	-	-	-
	CARD_NUMBER	Number	-	14	0	-	✓	-	-
	VAILD	Date	7	-	-	-	✓	-	-
	BKASH	Number	-	11	0	-	✓	-	-
	AMOUNT	Number	-	5	0	-	-	(0)	-
	C_ID	Number	-	3	0	-	✓	-	-
1 - 6									

Receive Method

create table Receive_method(Trans_ID number(7) primary key, Card_Number Number(14), Vaild date, Bkash number(11),Amount number(5) default(0));
Alter table Receive_method add(S_id Number(3));

The screenshot shows a web browser window with a SQL editor. The editor has a toolbar with 'Autocommit' checked, 'Display' set to 100, and 'Save' and 'Run' buttons. The SQL commands entered are:

```
create table Receive_method(Trans_ID number(7) primary key, Card_Number Number(14), Vaild date, Bkash number(11),Amount number(5) default(0));  
Alter table Receive_method add(S_id Number(3));  
  
describe Receive_method
```

Below the editor, the 'Results' tab is selected, showing the table structure for 'RECEIVE_METHOD'.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
RECEIVE_METHOD	TRANS_ID	Number	-	7	0	1	-	-	-
	CARD_NUMBER	Number	-	14	0	-	✓	-	-
	VAILD	Date	7	-	-	-	✓	-	-
	BKASH	Number	-	11	0	-	✓	-	-
	AMOUNT	Number	-	5	0	-	✓	(0)	-
	S_ID	Number	-	3	0	-	✓	-	-
1 - 6									

The bottom of the screen shows the 'Application Express 2.1.0.00.39' footer and a Windows taskbar with system information: 22°C Haze, 1:27 AM, and ENG language setting.

DATA INSERTION

Fruit Buisness

```
insert into Fruit_buisness(F_ID, F_name, Price) Values (01,'Mango', 170);
insert into Fruit_buisness(F_ID, F_name, Price) Values (02,'Apple', 340);
insert into Fruit_buisness(F_ID, F_name, Price) Values (03,'Jackfruit', 70);
insert into Fruit_buisness(F_ID, F_name, Price) Values (04,'Guava', 65);
insert into Fruit_buisness(F_ID, F_name, Price) Values (05,'Orange', 80);
```

The screenshot shows the Oracle Database Express Edition interface. The top navigation bar includes 'Home', 'SQL', and 'SQL Commands'. The main area displays the following SQL commands:

```
insert into Fruit_buisness(F_ID, F_name, Price) Values (01,'Mango', 170);
insert into Fruit_buisness(F_ID, F_name, Price) Values (02,'Apple', 340);
insert into Fruit_buisness(F_ID, F_name, Price) Values (03,'Jackfruit', 70);
insert into Fruit_buisness(F_ID, F_name, Price) Values (04,'Guava', 65);
insert into Fruit_buisness(F_ID, F_name, Price) Values (05,'Orange', 80);

select * from Fruit_buisness;
```

Below the commands, the 'Results' tab is active, showing a table with 5 rows:

F_ID	F_NAME	PRICE
1	Mango	170
2	Apple	340
3	Jackfruit	70
4	Guava	65
5	Orange	80

Below the table, it states '5 rows returned in 0.00 seconds' and provides a 'CSV Export' link. The bottom status bar shows 'Language: en-us' and 'Application Express 2.1.0.00.39 Copyright © 1999, 2006, Oracle. All rights reserved.'

Customer

```
insert into Customer (C_ID, C_name , Contact_Number) values (101,'Rahim',017111111111);
insert into Customer (C_ID, C_name , Contact_Number) values (102,'Karim',01712222222);
insert into Customer (C_ID, C_name , Contact_Number) values (103,'Judu',01733333333);
insert into Customer (C_ID, C_name , Contact_Number) values (104,'Kudu',01744444444);
insert into Customer (C_ID, C_name , Contact_Number) values (105,'Mohdu',01755555555);
```

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window contains the following commands:

```
insert into Customer (C_ID, C_name , Contact_Number) values (101,'Rahim',017111111111);
insert into Customer (C_ID, C_name , Contact_Number) values (102,'Karim',01712222222);
insert into Customer (C_ID, C_name , Contact_Number) values (103,'Judu',01733333333);
insert into Customer (C_ID, C_name , Contact_Number) values (104,'Kudu',01744444444);
insert into Customer (C_ID, C_name , Contact_Number) values (105,'Mohdu',01755555555);

select * from Customer;
```

The Results window displays the following data:

C_ID	C_NAME	CONTACT_NUMBER
101	Rahim	17111111111
105	Mohdu	17555555555
104	Kudu	17444444444
103	Judu	17333333333
102	Karim	17122222222

5 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39
Copyright © 1999, 2006, Oracle. All rights reserved.

Seller

```
insert into Seller (s_ID, s_name , Contact_Number) values (11,'Mr. X',018111111111);
insert into Seller (s_ID, s_name , Contact_Number) values (22,'Mr. Y',01512222222);
insert into Seller (s_ID, s_name , Contact_Number) values (33,'Mr. Z',01633333333);
insert into Seller (s_ID, s_name , Contact_Number) values (44,'Mr. K',01344444444);
insert into Seller (s_ID, s_name , Contact_Number) values (55,'Ms. X',01955555555);
```

The screenshot shows a web-based SQL interface. At the top, there's a browser window with multiple tabs. Below the browser, the page title is "Home > SQL > SQL Commands". The main area has a text input field with the following SQL commands:

```
insert into Seller (s_ID, s_name , Contact_Number) values (11,'Mr. X',018111111111);
insert into Seller (s_ID, s_name , Contact_Number) values (22,'Mr. Y',01512222222);
insert into Seller (s_ID, s_name , Contact_Number) values (33,'Mr. Z',01633333333);
insert into Seller (s_ID, s_name , Contact_Number) values (44,'Mr. K',01344444444);
insert into Seller (s_ID, s_name , Contact_Number) values (55,'Ms. X',01955555555);

select * from seller;
```

Below the input field, there are tabs for "Results", "Explain", "Describe", "Saved SQL", and "History". The "Results" tab is active, showing a table with 5 rows:

S_ID	S_NAME	CONTACT_NUMBER
11	Mr. X	18111111111
22	Mr. Y	15122222222
33	Mr. Z	16333333333
44	Mr. K	13444444444
55	Ms. X	19555555555

Below the table, it says "5 rows returned in 0.00 seconds" and there is a "CSV Export" link. At the bottom, there's a footer with "Application Express 2.1.0.00.39" and "Copyright © 1999, 2006, Oracle. All rights reserved.".

Payment_method

```
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(2621011, null, null, 0161321011, 570,101);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(9657455, 19705687412458,to_date('12-Apr-2026','DD-MM-YYYY'), null, 1010,102);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(7428901, 20226784146789,to_date('05-feb-2056','DD-MM-YYYY'), null, 5600,103);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(7564347, null, null, 01752094290, 340,104);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(6947310, null, null, 01915212290, 610,105);
```

The screenshot shows the Oracle Database Express Edition interface. The SQL Command window contains the following commands:

```
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(2621011, null, null, 0161321011, 570,101);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(9657455, 19705687412458,to_date('12-Apr-2026','DD-MM-YYYY'), null, 1010,102);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(7428901, 20226784146789,to_date('05-feb-2056','DD-MM-YYYY'), null, 5600,103);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(7564347, null, null, 01752094290, 340,104);
insert into Payment_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, C_ID) values(6947310, null, null, 01915212290, 610,105);

select * from payment_method
```

The Results window displays the following table:

TRANS_ID	CARD_NUMBER	VAILD	BKASH	AMOUNT	C_ID
2621011	-	-	161321011	570	101
9657455	19705687412458	12-APR-26	-	1010	102
7428901	20226784146789	05-FEB-56	-	5600	103
7564347	-	-	1752094290	340	104
6947310	-	-	1915212290	610	105

5 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39
Copyright © 1999-2006 Oracle. All rights reserved.
22°C Haze 2:11 AM

Receive_method

insert into Receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(2621011, 19705687416300,to_date('12-Dec-2026','DD-MM-YYYY'), null, 570,11);

insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, s_ID) values(9657455, 20005687412458,to_date('25-feb-2036','DD-MM-YYYY'), null, 1010,22);

insert into Receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(7428901, 20226784146789,to_date('05-may-2056','DD-MM-YYYY'), null, 5600,55);

insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(7564347, null, null, 01778573511, 340,33);

insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(6947310, null, null, 01587947638, 610,44);

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window contains the following commands:

```
insert into Receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(2621011, 19705687416300,to_date('12-Dec-2026','DD-MM-YYYY'), null, 570,11);
insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, s_ID) values(9657455, 20005687412458,to_date('25-feb-2036','DD-MM-YYYY'), null, 1010,22);
insert into Receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(7428901, 20226784146789,to_date('05-may-2056','DD-MM-YYYY'), null, 5600,55);
insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(7564347, null, null, 01778573511, 340,33);
insert into receive_method(Trans_ID , Card_Number , Vaild, Bkash,Amount, S_ID) values(6947310, null, null, 01587947638, 610,44);

select * from receive_method
```

The Results tab shows the following data:

TRANS_ID	CARD_NUMBER	VAILD	BKASH	AMOUNT	S_ID
2621011	19705687416300	12-DEC-26	-	570	11
9657455	20005687412458	25-FEB-36	-	1010	22
7428901	20226784146789	05-MAY-56	-	5600	55
7564347	-	-	1778573511	340	33
6947310	-	-	1587947638	610	44

5 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39
Copyright © 1999, 2006 Oracle. All rights reserved.
22°C Haze 2:26 AM

Q. Display all information between Customer and Payment Method

Ans: select C.C_ID , C.C_name, C.Contact_Number, P.Trans_ID, P.Amount from Customer C , Payment_method P where C.C_ID = P.C_ID;

The screenshot shows the Oracle Database Express Edition web interface. The user is logged in as 'SYS'. The breadcrumb navigation is 'Home > SQL > SQL Commands'. The 'Autocommit' checkbox is checked, and the 'Display' dropdown is set to '100'. The 'Save' and 'Run' buttons are visible. The SQL command entered is:

```
select C.C_ID , C.C_name, C.Contact_Number, P.Trans_ID, P.Amount from Customer C , Payment_method P where C.C_ID = P.C_ID;
```

Below the command, the 'describe payment_method' command is also entered. The 'Results' tab is selected, showing a table with 5 rows and 5 columns: C_ID, C_NAME, CONTACT_NUMBER, TRANS_ID, and AMOUNT.

C_ID	C_NAME	CONTACT_NUMBER	TRANS_ID	AMOUNT
101	Rahim	1711111111	2621011	570
102	Karim	1712222222	9657455	1010
103	Judu	1733333333	7428901	5600
104	Kudu	1744444444	7564347	340
105	Mohdu	1755555555	6947310	610

5 rows returned in 0.00 seconds. CSV Export

Q. Display all information between Customer and Seller

Ans: select C.C_ID "CUSTOMER ID" , C.C_name "CUSTOMER",S.S_name "RECEVIER",S.S_ID "SELLER ID", P.Trans_ID, P.Amount from Customer C , Payment_method P ,Seller S , Receive_method R where C.C_ID = P.C_ID AND P.Trans_ID = R.Trans_ID AND S.S_ID =R.S_ID;

The screenshot shows the Oracle Database Express Edition SQL Commands interface. The query entered is:

```
select C.C_ID "CUSTOMER ID" , C.C_name "CUSTOMER",S.S_name "RECEVIER",S.S_ID "SELLER ID", P.Trans_ID, P.Amount from Customer C , Payment_method P ,Seller S , Receive_method R where C.C_ID = P.C_ID AND P.Trans_ID = R.Trans_ID AND S.S_ID =R.S_ID;
```

The results are displayed in a table with 6 columns: CUSTOMER ID, CUSTOMER, RECEVIER, SELLER ID, TRANS_ID, and AMOUNT. There are 5 rows of data.

CUSTOMER ID	CUSTOMER	RECEVIER	SELLER ID	TRANS_ID	AMOUNT
101	Rahim	Mr. X	11	2621011	570
102	Karim	Mr. Y	22	9657455	1010
104	Kudu	Mr. Z	33	7564347	340
105	Mohdu	Mr. K	44	6947310	610
103	Judu	Ms. X	55	7428901	5600

5 rows returned in 0.00 seconds

Outer join 1

Q. Display all information between customer and payment method

Ans: select c.c_id,c.c_name,p.trans_id,p.amount
from customer c, payment_method p
where c.c_id(+)=p.trans_id;

The screenshot shows a web browser window with multiple tabs. The active tab is 'SQL Commands'. The browser address bar shows '127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::'. The page header indicates 'User: SYS' and 'Home > SQL > SQL Commands'. Below the header, there is a toolbar with 'Autocommit' checked, 'Display' set to '100', and 'Save' and 'Run' buttons. The main area contains the SQL query:

```
select c.c_id,c.c_name,p.trans_id,p.amount
from customer c, payment_method p
where c.c_id(+)=p.trans_id;
```

 Below the query, there is a 'Results' tab selected, showing a table with 5 rows. The table has columns: C_ID, C_NAME, TRANS_ID, and AMOUNT. The data rows are: (-, -, 6947310, 610), (-, -, 2621011, 570), (-, -, 7428901, 5600), (-, -, 9657455, 1010), and (-, -, 7564347, 340). Below the table, it says '5 rows returned in 0.00 seconds' and 'CSV Export'. The bottom of the page shows 'Application Express 2.1.0.00.39' and a Windows taskbar with system tray information: '24°C Sunny', '11:40 AM', and 'ENG'.

User: SYS

Home > SQL > SQL Commands

☒ Autocommit Display 100 Save Run

```
select c.c_id,c.c_name,p.trans_id,p.amount
from customer c, payment_method p
where c.c_id(+)=p.trans_id;
```

Results Explain Describe Saved SQL History

C_ID	C_NAME	TRANS_ID	AMOUNT
-	-	6947310	610
-	-	2621011	570
-	-	7428901	5600
-	-	9657455	1010
-	-	7564347	340

5 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39

24°C Sunny 11:40 AM ENG

Q. Display all information between Customer and Seller

Ans:
`select c.c_id,c.c_name,s.s_name,s.s_id`
`from customer c,seller s`
`where c.c_id(+) =s.s_id;`

The screenshot shows a web browser window with multiple tabs. The active tab is 'SQL Commands'. The address bar shows a URL starting with '127.0.0.1:8080/apex/'. The page content includes a header 'User: SYS' and a breadcrumb 'Home > SQL > SQL Commands'. Below this is a form with a checked 'Autocommit' checkbox, a 'Display' dropdown set to '100', and 'Save' and 'Run' buttons. The SQL query entered is:

```
select c.c_id,c.c_name,s.s_name,s.s_id
from customer c,seller s
where c.c_id(+) =s.s_id;
```

Below the query editor, there are tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History'. The 'Results' tab is active, showing a table with 5 rows and 4 columns: C_ID, C_NAME, S_NAME, and S_ID.

C_ID	C_NAME	S_NAME	S_ID
-	-	Mr. X	11
-	-	Ms. X	55
-	-	Mr. K	44
-	-	Mr. Z	33
-	-	Mr. Y	22

Below the table, it says '5 rows returned in 0.00 seconds' and there is a 'CSV Export' link. The bottom of the page shows 'Application Express 2.1.0.00.39'.

Q. Display Self-Join of Payment Method Table

Ans: select A.Trans_ID,A.CARD_NUMBER,A.VAILD ,A.amount,B.Trans_ID,B.BKASH
from payment_method A,payment_method B
where A.Trans_ID = B.Trans_ID ;

The screenshot shows a web-based SQL interface with the following components:

- Browser Tabs:** SQL Commands, Untitled Diagram.drawio - diagr..., Sign in - GitLab, Ekhn Onek Raat | এখন অনেক..., transaction - Google Search, (1) Facebook.
- Address Bar:** 127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::
- User:** SYS
- Navigation:** Home > SQL > SQL Commands
- Query Editor:**
 - Autocommit: ☒ Display: 100
 - Buttons: Save, Run
 - SQL Query:

```
select A.Trans_ID,A.CARD_NUMBER,A.VAILD ,A.amount,B.Trans_ID,B.BKASH
from payment_method A,payment_method B
where A.Trans_ID = B.Trans_ID ;
```
- Results Tab:**
 - Buttons: Results (selected), Explain, Describe, Saved SQL, History
 - Table with 6 columns: TRANS_ID, CARD_NUMBER, VAILD, AMOUNT, TRANS_ID, BKASH
 - 5 rows of data.
 - Footer: 5 rows returned in 0.00 seconds, [CSV Export](#)
- Footer:** Application Express 2.1.0.00.39, Copyright © 1999, 2006, Oracle. All rights reserved.

TRANS_ID	CARD_NUMBER	VAILD	AMOUNT	TRANS_ID	BKASH
2621011	-	-	570	2621011	161321011
9657455	19705687412458	12-APR-26	1010	9657455	-
7428901	20226784146789	05-FEB-56	5600	7428901	-
7564347	-	-	340	7564347	1752094290
6947310	-	-	610	6947310	1915212290

Q. Display Self-Join of Payment Method Table

Ans: select A.Trans_ID,A.CARD_NUMBER,A.VAILD ,A.amount,B.Trans_ID,B.BKASH
from payment_method A,Receive_method B
where A.Amount>B.Amount;

The screenshot shows a web-based SQL execution environment. The query entered is:

```
select A.Trans_ID,A.CARD_NUMBER,A.VAILD ,A.amount,B.Trans_ID,B.BKASH
from payment_method A,Receive_method B
where A.Amount>B.Amount;
```

The results are displayed in a table with 6 columns: TRANS_ID, CARD_NUMBER, VAILD, AMOUNT, TRANS_ID, and BKASH. There are 10 rows of data returned.

TRANS_ID	CARD_NUMBER	VAILD	AMOUNT	TRANS_ID	BKASH
2621011	-	-	570	7564347	1778573511
9657455	19705687412458	12-APR-26	1010	2621011	-
9657455	19705687412458	12-APR-26	1010	7564347	1778573511
9657455	19705687412458	12-APR-26	1010	6947310	1587947638
7428901	20226784146789	05-FEB-56	5600	2621011	-
7428901	20226784146789	05-FEB-56	5600	9657455	-
7428901	20226784146789	05-FEB-56	5600	7564347	1778573511
7428901	20226784146789	05-FEB-56	5600	6947310	1587947638
6947310	-	-	610	2621011	-
6947310	-	-	610	7564347	1778573511

10 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39
Copyright © 1999-2006 Oracle. All rights reserved.
24°C Haze

SUBQUERY NO 1

Qus: Display all who pays maximum amount of money

Ans: select * from payment_method

SQL Commands

Home > SQL > SQL Commands

☒ Autocommit Display 100 Save Run

```
select * from payment_method
where amount = (select MAX(amount) from payment_method);
```

Results Explain Describe Saved SQL History

TRANS_ID	CARD_NUMBER	VAILD	BKASH	AMOUNT	C_ID
7428901	20226784146789	05-FEB-56	-	5600	103

1 rows returned in 0.02 seconds [CSV Export](#)

Application Express 2.1.0.00.39

Language: en-us Copyright © 1999, 2006, Oracle. All rights reserved.

SubQuery NO. 2

QUS: Display all information who receive 2nd Minimum amount of money

Ans: select* from receive_method

where amount =(select MIN(amount) from receive_method

where Amount > (select MIN(AMOUNT) from receive_method));

Home > SQL > SQL Commands

☒ Autocommit Display 100 Save Run

```
select* from receive_method
where amount =(select MIN(amount) from receive_method
where Amount > (select MIN(AMOUNT) from receive_method));

describe receive_method
```

Results Explain Describe Saved SQL History

TRANS_ID	CARD_NUMBER	VAILD	BKASH	AMOUNT	S_ID
2621011	19705687416300	12-DEC-26	-	570	11

1 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39
Copyright © 1999, 2006, Oracle. All rights reserved.

Language: en-us

SubQuery N0. 3

QUS: Display all information who receive maximum amount of money

Ans: select* from receive_method

where amount =(select MAX(Amount) from receive_method);

The screenshot shows the Oracle Database Express Edition web interface. The browser tabs include 'SQL Commands', 'Untitled Diagram.drawio', 'Sign in - GitLab', 'Ekhon Onek Raat', 'transaction - Google Search', and '(1) Facebook'. The address bar shows a local URL. The page header includes the Oracle logo and 'Database Express Edition'. The user is logged in as 'SYS'. The breadcrumb navigation is 'Home > SQL > SQL Commands'. The SQL editor has 'Autocommit' checked and 'Display' set to '100'. The SQL query entered is:

```
select *  
from receive_method  
where Amount =(select MAX(amount)  
                from receive_method);
```

 The 'Run' button is visible. Below the editor, the 'Results' tab is selected, showing a table with 6 columns: TRANS_ID, CARD_NUMBER, VAILD, BKASH, AMOUNT, and S_ID. The table contains one row of data. Below the table, it says '1 rows returned in 0.00 seconds' and provides a 'CSV Export' link. The footer shows 'Application Express 2.1.0.00.39' and a copyright notice.

ORACLE Database Express Edition

User: SYS

Home > SQL > SQL Commands

☒ Autocommit Display 100 Save Run

```
select *  
from receive_method  
where Amount =(select MAX(amount)  
                from receive_method);
```

Results Explain Describe Saved SQL History

TRANS_ID	CARD_NUMBER	VAILD	BKASH	AMOUNT	S_ID
7428901	20226784146789	05-MAY-56	-	5600	55

1 rows returned in 0.00 seconds [CSV Export](#)

Application Express 2.1.0.00.39

Copyright © 1996-2006 Oracle. All rights reserved.

SubQuery NO. 4

QUS: Display all information who pays 2nd maximum amount of money

Ans: select* from receive_method

where amount =(select MIN(amount) from receive_method

where Amount > (select MIN(AMOUNT) from receive_method));

The screenshot shows the Oracle Database Express Edition web interface. The browser tabs include 'SQL Commands', 'Untitled Diagram.drawio - drawio', 'Sign in - GitLab', 'Ekhon Onek Raat | এখন অনেক রাত', 'transaction - Google Search', and '(1) Facebook'. The address bar shows '127.0.0.1:8080/apex/?p=4500:1003:2241271529133695::NO...'. The Oracle logo and 'Database Express Edition' are visible. The user is logged in as 'SYS'. The breadcrumb navigation is 'Home > SQL > SQL Commands'. In the SQL Commands window, the 'Autocommit' checkbox is checked, and the 'Display' dropdown is set to '100'. The SQL query entered is:

```
select *  
from payment_method  
where amount=(select MAX(amount) |  
               from payment_method  
               where Amount>(select MAX(Amount)  
                             from payment_method));  
  
describe payment_method
```

Below the query window, there are 'Save' and 'Run' buttons. The 'Results' tab is selected, showing 'no data found'. The footer of the application shows 'Application Express 2.1.0.00.39' and 'Copyright © 1999, 2006 Oracle. All rights reserved.'.

SIMPLE VIEW

Q. Create a view of Transaction ID , Amount and show the table where Amount is not null

A. create view payment_vu1 ("transaction","Payment_Amount") as select Trans_id, amount from
payment_method
where amount is not null;
select * from payment_vu1

The screenshot shows a web browser window with a SQL command interface. The browser's address bar shows a URL starting with 127.0.0.1. The page has a navigation bar with "Home > SQL > SQL Commands". Below this, there's a toolbar with "Autocommit" (checked), "Display" (set to 100), "Save", and "Run" buttons. The main text area contains the following SQL code:

```
create view payment_vu1 ("transaction","Payment_Amount") as select Trans_id, amount from  
payment_method  
where amount is not null;  
  
select * from payment_vu1
```

Below the code area, there are tabs for "Results", "Explain", "Describe", "Saved SQL", and "History". The "Results" tab is active, displaying a table with 5 rows of data:

Transaction	Payment_Amount
2621011	570
9657455	1010
7428901	5600
7564347	340
6947310	610

At the bottom of the results section, it says "5 rows returned in 0.00 seconds" and provides a "CSV Export" link. The browser's taskbar at the bottom shows the system time as 3:15 AM and the temperature as 21°C.

Complex view

Q. Create a Complex view of Customer Name , ID , Amount of payment , Transaction ID

Ans: create View Paid (Name, Customer_ID,Taka,
transaction ID) AS Select C.C name,c.C ID,P.Amount,P.Trans ID from Customer C , Payment method P where C.C ID = P.C_ID;

SQL Commands x Untitled Diagram.drawio - diagra x Sign in - GitLab x Ekhn Onek Raat | এখন অনেক রাত x transaction - Google Search x Downloads x +

127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::

Gmail pirate Govt Scholar YouTube LinkedIn Profile Estiyak-rubs (Estiya... Home | LightOJ Navigation - Algorit... Codeforces Stoptalks কন্সিউটার প্রোগ্রা... speedtest.googlefib... chrome pass free do... Google Translate Maps SAT Subject Tests |... >>

☒ Autocommit Display 100 Save Run

```
create View Paid (Name, Customer_ID,Taka,  
transaction_ID) AS Select C.C_name,c.C_ID,P.Amount,P.Trans_ID from Customer C , Payment_method P  
where C.C_ID = P.C_ID;  
  
select* from Paid;
```

 **3**

Results Explain Describe Saved SQL History

NAME	CUSTOMER_ID	TAKA	TRANSACTION_ID
Rahim	101	570	2621011
Karim	102	1010	9657455
Judu	103	5600	7428901
Kudu	104	340	7564347
Mohdu	105	610	6947310

5 rows returned in 0.02 seconds

[CSV Export](#)

Application Express 2.1.0.00.39

Language: en-us

Copyright © 1999. 2006. Oracle. All rights reserved.

Constraint

Payment_method

Alter table payment_method modify amount not null;

SQL Commands

Untitled Diagram.drawio - diagra

Sign in - GitLab

The Local Train - Aalas Ka Pedh -

+

127.0.0.1:8080/apex/f?p=4500:1003:2241271529133695::NO::

Gmail pirate Govt Scholar YouTube LinkedIn Profile Estiyak-rubs (Estiya... Home | LightOJ Navigation - Algorit... Codeforces Stoptalks কম্পিউটার প্রোগ্রা... speedtest.googlefib... chrome pass free do... Google Translate Maps SAT Subject Tests |...

Home Logout Help

User: SYS

Home > SQL > SQL Commands

☒ Autocommit Display 100

Save Run

Alter table payment_method modify amount not null;

describe payment_method

Results Explain Describe Saved SQL History

Object Type TABLE Object PAYMENT_METHOD

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT_METHOD	TRANS_ID	Number	-	7	0	1	-	-	-
	CARD_NUMBER	Number	-	14	0	-	✓	-	-
	VAILD	Date	7	-	-	-	✓	-	-
	BKASH	Number	-	11	0	-	✓	-	-
	AMOUNT	Number	-	5	0	-	-	(0)	-
	C_ID	Number	-	3	0	-	✓	-	-
1 - 6									

22°C Haze 2:45 AM